

BullSO 2026 Invitational Disease Detectives B Answer Key

Multiple Choice (1 points each) - 76 points

1. A clinician diagnoses tuberculosis in a patient and prescribes treatment. A public health professional investigates housing conditions and contacts of the patient. This contrast best illustrates:

- A. Primary vs secondary prevention
- B. Individual vs population-based approaches**
- C. Incidence vs prevalence
- D. Surveillance vs outbreak investigation

2. Epidemiology differs from clinical medicine primarily because epidemiology:

- A. Relies on laboratory confirmation
- B. Focuses on treatment outcomes
- C. Studies health events in populations**
- D. Addresses only infectious diseases

3. Which historical development most directly contributed to modern epidemiologic surveillance?

- A. Germ theory
- B. Vaccination
- C. Routine collection of vital statistics**
- D. Antibiotic therapy

4. John Snow's cholera investigation is considered foundational because it demonstrated:

- A. The effectiveness of quarantine
- B. The importance of randomized trials
- C. Use of descriptive epidemiology to identify a source**
- D. The role of immunity in disease spread

5. Which of the following is not a core role of epidemiology in public health?

- A. Identifying risk factors
- B. Evaluating interventions
- C. Establishing clinical diagnoses**
- D. Guiding prevention strategies

6. The spectrum of disease refers to:

- A. Genetic diversity of pathogens
- B. Range of disease severity from subclinical to fatal**
- C. Environmental variation in exposure
- D. Differences between chronic and acute disease

7. A disease that progresses from exposure → latency → clinical disease → outcome without intervention is demonstrating:
- A. Herd immunity
 - B. Chain of transmission
 - C. Natural history of disease
 - D. Ecological fallacy
8. Which pair correctly matches the disease category with an example?
- A. Noninfectious – malaria
 - B. Viral – tuberculosis
 - C. Prion – Creutzfeldt-Jakob disease
 - D. Bacterial – influenza
9. Which noninfectious cause of disease is most closely associated with dose-response effects?
- A. Genetic mutation
 - B. Toxic chemical exposure
 - C. Congenital anomaly
 - D. Autoimmune disease
10. A risk is best defined as:
- A. The chance of receiving a diagnosis
 - B. The probability of developing disease
 - C. The presence of exposure
 - D. The severity of illness compared to others
11. Which situation best represents an epidemic rather than an outbreak?
- A. Two linked cases of botulism
 - B. Expected seasonal influenza cases
 - C. Sudden regional increase in measles cases
 - D. Endemic malaria cases
12. Disease that is consistently present at a baseline level within a population is described as:
- A. Epidemic
 - B. Pandemic
 - C. Sporadic
 - D. Endemic
13. Which is an example of active surveillance?
- A. Physicians voluntarily reporting cases
 - B. Automated lab reporting
 - C. Health department actively contacting hospitals
 - D. Reviewing death certificates
14. The primary purpose of public health surveillance is to:

- A. Assign individual risk
- B. Detect and respond to health problems
- C. Prove causation
- D. Replace epidemiologic studies

15. Which is not one of the five steps of surveillance?

- A. Data collection
- B. Data analysis
- C. Hypothesis testing
- D. Dissemination of findings

16. A surveillance system that is timely but incomplete most likely sacrifices:

- A. Sensitivity
- B. Representativeness
- C. Flexibility
- D. Simplicity

17. Which attribute of a surveillance system reflects its ability to correctly identify true cases?

- A. Predictive value positive
- B. Sensitivity
- C. Stability
- D. Acceptability

18. Syndromic surveillance is particularly useful for:

- A. Confirming diagnoses
- B. Early outbreak detection
- C. Measuring disease prevalence
- D. Determining causation

19. Which data source is most appropriate for monitoring long-term mortality trends?

- A. Case-control studies
- B. Hospital discharge data
- C. Vital statistics records
- D. Sentinel surveillance

20. A limitation common to passive surveillance systems is:

- A. High cost
- B. Overreporting
- C. Underreporting
- D. Excessive timeliness

21. The first operational step in an outbreak investigation is to:

- A. Implement control measures
- B. Verify the diagnosis

- C. Generate hypotheses
 - D. Describe cases by time, place, and person
22. Developing a case definition primarily helps investigators to:
- A. Establish causality
 - B. Identify exposure sources
 - C. Standardize case identification
 - D. Calculate risk ratios
23. Which component is not required in a standard case definition?
- A. Clinical criteria
 - B. Laboratory criteria
 - C. Person, place, and time
 - D. Proposed intervention
24. An epidemic curve with a sharp rise and gradual decline suggests:
- A. Continuous common-source exposure
 - B. Point-source exposure
 - C. Propagated transmission
 - D. Endemic transmission
25. A propagated outbreak is most likely associated with:
- A. Foodborne illness
 - B. Water contamination
 - C. Person-to-person transmission
 - D. Chemical exposure
26. Line listings are particularly useful for:
- A. Calculating incidence rates
 - B. Summarizing individual case characteristics
 - C. Identifying causal mechanisms
 - D. Testing hypotheses statistically
27. Which study design is most efficient for investigating a rare disease?
- A. Cohort study
 - B. Case-control study
 - C. Cross-sectional study
 - D. Ecological study
28. A major limitation of ecological studies is:
- A. High cost
 - B. Recall bias
 - C. Ecological fallacy
 - D. Small sample size

29. Which study design measures exposure and outcome at the same time?

- A. Cohort
- B. Case-control
- C. Cross-sectional
- D. Experimental

30. In a cohort study, the primary measure of association is:

- A. Odds ratio
- B. Risk difference
- C. Relative risk
- D. Prevalence ratio

31. Which measure is most appropriate for a case-control study?

- A. Incidence rate
- B. Relative risk
- C. Odds ratio
- D. Attack rate

32. An odds ratio of 1.0 indicates:

- A. Strong association
- B. Protective exposure
- C. No association
- D. Causal relationship

33. Which triad best represents disease causation in epidemiology?

- A. Exposure–Outcome–Bias
- B. Agent–Host–Environment
- C. Time–Place–Person
- D. Susceptible–Infected–Recovered

34. The chain of transmission includes all of the following except:

- A. Infectious agent
- B. Reservoir
- C. Vector
- D. Incubation period

35. PFGE and SNP mapping are primarily used to:

- A. Diagnose individual cases
- B. Measure disease prevalence
- C. Determine genetic relatedness of pathogens
- D. Calculate attack rates

36. PulseNet is best described as:

- A. A surveillance database for chronic disease
- B. A national molecular subtyping network

- C. A vaccine registry
- D. An international outbreak response team

37. Attack rate is best defined as:

- A. Prevalence of disease
- B. Incidence proportion during an outbreak
- C. Incidence rate over time
- D. Case-fatality ratio

38. A relative risk greater than 1 indicates that the exposure:

- A. Is protective
- B. Has no effect
- C. Is associated with increased risk
- D. Proves causation

39. Confidence intervals are useful because they:

- A. Eliminate bias
- B. Measure statistical power
- C. Indicate precision of estimates
- D. Replace hypothesis testing

40. A confidence interval that includes 1.0 for a relative risk suggests:

- A. Strong association
- B. Statistical significance
- C. No statistically significant association
- D. Measurement error

41. Tiebreaker!

Confounding occurs when:

- A. Random error affects results
- B. An exposure is misclassified
- C. A third factor is associated with both exposure and outcome
- D. Sample size is too small

42. Stratification is primarily used to:

- A. Increase sample size
- B. Control confounding
- C. Improve surveillance sensitivity
- D. Detect outbreaks earlier

43. Which bias is most common in case-control studies?

- A. Selection bias
- B. Recall bias
- C. Measurement bias
- D. Publication bias

44. Vaccine efficacy is typically measured using:
- A. Observational studies
 - B. Case-control studies
 - C. Randomized controlled trials
 - D. Ecological studies
45. Vaccine effectiveness differs from efficacy because it measures:
- A. Immune response
 - B. Biological plausibility
 - C. Real-world performance
 - D. Herd immunity threshold
46. Incidence rate differs from incidence proportion because it incorporates:
- A. Disease severity
 - B. Population size
 - C. Time at risk
 - D. Case fatality
47. Prevalence is most influenced by:
- A. Incidence and disease duration
 - B. Attack rate
 - C. Mortality rate only
 - D. Exposure frequency
48. Mortality rate measures:
- A. Risk of death among cases
 - B. Deaths in a population over time
 - C. Proportion of fatal cases
 - D. Life expectancy
49. Case-fatality ratio is best described as:
- A. Deaths divided by population
 - B. Deaths among diagnosed cases
 - C. Incidence of fatal disease
 - D. Mortality over time
50. A rising prevalence with stable incidence most likely indicates:
- A. Increased transmission
 - B. Improved case detection
 - C. Longer survival
 - D. Reporting bias
51. Which graph is most appropriate for showing disease trends over time?
- A. Pie chart

- B. Bar graph
- C. Line graph
- D. Scatter plot

52. Descriptive epidemiology primarily addresses:

- A. Why disease occurs
- B. How interventions work
- C. Who, where, and when
- D. Biological mechanisms

53. Primary prevention focuses on:

- A. Early detection
- B. Reducing disease complications
- C. Preventing disease occurrence
- D. Rehabilitation

54. Secondary prevention is best illustrated by:

- A. Vaccination
- B. Screening tests
- C. Physical therapy
- D. Palliative care

55. Tertiary prevention aims to:

- A. Prevent exposure
- B. Detect disease early
- C. Reduce disability
- D. Eliminate risk factors

56. Herd immunity protects individuals by:

- A. Eliminating pathogens
- B. Reducing opportunities for transmission
- C. Increasing individual resistance
- D. Preventing mutation

57. The basic reproductive number (R_0) represents:

- A. Case fatality rate
- B. Average secondary cases in a susceptible population
- C. Duration of infection
- D. Vaccine effectiveness

58. Herd immunity threshold is calculated using:

- A. $1 - R_0$
- B. $R_0 \div \text{population}$
- C. $1 - (1/R_0)$
- D. $R_0 \times \text{incidence}$

59. An effective reproductive number (R_e) less than 1 indicates:

- A. Endemic transmission
- B. Growing outbreak
- C. Declining transmission
- D. No exposure

60. Identifying trends, calculating rates, and comparing groups are core components of:

- A. Clinical medicine
- B. Laboratory science
- C. Epidemiologic analysis
- D. Health policy

61. **Tiebreaker!**

In an outbreak of Legionnaires' disease linked to a hotel cooling tower, which labeling of the epidemiologic triangle best represents this scenario?

- A.
Agent: *Legionella pneumophila*
Host: Elderly hotel guests
Environment: Aerosolized water system
- B.
Agent: Cooling tower
Host: Hotel guests
Environment: Bacteria
- C.
Agent: Aerosols
Host: Water system
Environment: Elderly guests
- D.
Agent: Hotel
Host: Bacteria
Environment: Air

62. A city implements syndromic surveillance using emergency department chief complaints.

The system identifies a sharp increase in "flu-like illness," but laboratory-confirmed cases remain unchanged.

Which is the MOST likely explanation?

- A. The system has low sensitivity
- B. The system has high specificity
- C. The disease incubation period has shortened
- D. Changes in healthcare-seeking behavior are inflating case counts

63) A clinician treats a patient with severe dehydration following a gastrointestinal illness. A public health epidemiologist reviews the same case.

Which action BEST reflects the public health approach?

- A. Prescribing intravenous fluids
- B. Ordering diagnostic stool cultures
- C. Monitoring electrolyte balance
- D. Interviewing recent contacts to identify a common exposure

64) Which surveillance system attribute is MOST important when the goal is early outbreak detection, even at the cost of false positives?

- A. Specificity
- B. Representativeness
- C. Sensitivity
- D. Predictive value positive

65) An investigator wants to examine whether attendance at a one-day event is associated with illness. Exposure status is known for all attendees.

Which study design is MOST appropriate?

- A. Case-control
- B. Cohort
- C. Cross-sectional
- D. Ecological

66) Which situation BEST illustrates recall bias?

- A. Laboratory tests misclassify cases
- B. Cases are more likely than controls to remember exposures
- C. Health records omit mild cases
- D. An exposure is associated with age

67) An apparent association is observed between ice cream consumption and drowning deaths.

Which variable is MOST likely a confounder?

- A. Age
- B. Temperature

- C. Gender
- D. Swimming ability

68) An epi curve shows a sharp rise and fall in cases over 24 hours, with no secondary peaks.

This pattern MOST strongly suggests:

- A. Propagated outbreak
- B. Continuous common source
- C. Point-source exposure
- D. Seasonal epidemic

69) If two groups have identical attack rates, the relative risk will be:

- A. 0
- B. 1
- C. Greater than 1
- D. Undefined

70) A relative risk of 0.5 indicates that exposed individuals:

- A. Have half the risk of disease compared to unexposed
- B. Are twice as likely to become ill
- C. Have no association with disease
- D. Have a protective exposure that proves causality

71) In which situation is an odds ratio preferred over a relative risk?

- A. When disease incidence is high
- B. When studying rare diseases
- C. When exposure is common
- D. When conducting cohort studies

72) Which Bradford Hill criterion is satisfied if exposure ALWAYS occurs before disease onset?

- A. Consistency
- B. Strength
- C. Temporality
- D. Plausibility

73) Which scenario MOST weakens a causal claim?

- A. Lack of temporal relationship
- B. Strong association
- C. Biological plausibility
- D. Dose-response relationship

74) Which intervention is an example of secondary prevention?

- A. Vaccination campaign
- B. Environmental sanitation
- C. Screening asymptomatic individuals
- D. Rehabilitation services

75) Which situation raises the GREATEST ethical concern in surveillance?

- A. Use of aggregated data
- B. Delayed reporting
- C. Low sensitivity
- D. Collection of identifiable personal data without consent

76) After analyzing outbreak data, no significant association is found between suspected exposure and illness.

What is the BEST next step?

- A. Conclude there is no outbreak
- B. Revise the hypothesis
- C. Stop the investigation
- D. Increase sample size without changing approach

Matching (1 points each) - 20 points

1. Surveillance L	A. A set of considerations used to evaluate whether an observed association is likely causal
2. Outbreak E	B. Any systematic error that leads to an incorrect estimate of association
3. Epidemic T	C. The proportion of a population that becomes ill after exposure during a specified time period
4. Pandemic Q	D. A biological organism that transmits a pathogen between hosts
5. Attack Rate C	E. A sudden increase in disease cases above expected levels in a specific population
6. Incidence Rate O	F. A detailed description of who qualifies as a case in an investigation
7. Prevalence I	G. A variable associated with both the exposure and outcome that distorts the true relationship
8. Relative Risk H	H. A measure comparing the probability of disease in exposed versus unexposed groups
9. Odds Ratio R	I. The total number of existing cases in a population at a given time
10. Case Definition F	J. Measures taken to prevent disease before it occurs
11. Confounding Variable G	K. An outbreak caused by person-to-person transmission
12. Bias B	L. Continuous, systematic collection, analysis, and interpretation of health data
13. Point-Source Outbreak M	M. An outbreak limited to a single exposure source over a short time
14. Propagated Outbreak K	N. Measures aimed at early detection and prompt treatment
15. Vector D	O. The rate at which new cases occur in a population over a specified period
16. Host S	P. The physical and social factors that influence disease transmission
17. Environment P	Q. A disease occurrence affecting many countries or continents
18. Primary Prevention J	R. A measure of association commonly used in case-control studies
19. Secondary Prevention N	S. An organism that can become infected by a disease agent
20. Bradford Hill Criteria A	T. A disease occurrence greater than expected within a region or population

Open Ended - 54 points

1) A city health department notices an increase in emergency room visits for respiratory symptoms after a heatwave.

Explain two reasons why this observation alone does not confirm an outbreak. (4 points)

Seasonal variation; increased healthcare utilization; lack of baseline comparison

2) A surveillance system detects cases rapidly but frequently misclassifies healthy individuals as cases.

a) Identify two surveillance attributes illustrated. (4 points)

High sensitivity, low specificity

b) Explain how this tradeoff affects public health decision-making. (3 points)

leads to false alarms but early detection

3) Compare the clinical approach and public health approach to disease using the same illness, explaining how conclusions and interventions might differ. (4 points)

Clinical treats individuals; public health focuses on exposure control

4) Following a 3-day outdoor festival, 180 cases of neurological symptoms are reported. Symptoms include dizziness, confusion, and tremors.

Exposure	Cases	Non-Cases	Total
Drank Well Water	120	80	200
Did Not	60	140	200

a) Calculate the attack rate for exposed and unexposed groups. (4 points)

Exposed: $120/200 = 60\%$

Unexposed: $60/200 = 30\%$

b) Calculate the relative risk and interpret it. (4 points)

$RR = 0.60 / 0.30 = 2.0$

Exposed individuals twice as likely to be ill

c) Based on these data alone, does the evidence support a causal relationship? Explain two limitations. (4 points)

No temporal proof; possible confounding; exposure misclassification

d) Identify one potential confounder and explain how it could bias results. (3 points)

Confounder: food consumption, duration at festival

5) An epi curve shows:

A gradual rise in cases

Multiple peaks over 10 days

Overlapping exposure periods

a) Identify the most likely mode of transmission. (3 points)

Propagated outbreak

b) Explain how this pattern guides hypothesis generation. (3 points)

Suggests person-to-person or ongoing exposure

c) Identify two steps of outbreak investigation that should occur next and justify them. (6 points)

Verify diagnosis; refine case definition; environmental sampling

6) A population of 50,000 reports:

1,200 existing cases on January 1

300 new cases over the year

75 disease-specific deaths

Calculate:

a) Point prevalence (3 points)

$$1,200 / 50,000 = 2.4\%$$

b) Incidence proportion (3 points)

$$300 / 48,800 \approx 0.61\%$$

c) Mortality rate (3 points)

$$75 / 50,000 = 0.15\%$$

d) Explain one reason prevalence may increase even when incidence is decreasing. (3 points)

Longer survival or chronic disease